

MAL-SME-Banking

Architecture diagram. Seven views: the legend, the logical reference architecture, the Decision Engine, Build versus Buy, money flow on a Murabaha-funded invoice, the Sharia lifecycle, and the deployment topology.

Customer

UAE SMEs of 5 to 50 employees, mainland and free zone.

Licensing path

ADGM FSP Category 2 plus Category 4, with optional RegLab sandbox.

Sharia governance

AAOIFI-aligned, internal SSB, audited quarterly.

Cloud

AWS me-central-1 in Abu Dhabi, DR in me-south-1.

Legend

Layer colours used across every diagram on this page. Each layer has its own ownership, deployment cadence, and SLO.

☐ Client experience iOS · Android · Web · Ops Console	☐ Edge & API gateway CDN · WAF · OIDC · Rate limit
☐ Domain services Customer · Cards · Pay · Lend · FX · ...	☐ Core banking platform Mambu · IBAN issuer · Ledger
☐ Data platform OLTP · Streaming · Lakehouse · Feature store	☐ External integrations Rails · KYC · Bureau · Accounting · FX
☐ AI / ML platform Fraud · Forecast · Underwriting · LLM	☐ Sharia compliance SSB · Murabaha · Halal screen · Zakat
☐ Regulatory & compliance CBUAE reporting · goAML · audit log	☐ Operations & observability SRE · CS · Treasury ops · DataDog

§ 02 · Primary view

Logical reference architecture

Top-down stack: customer apps flow through an API gateway into domain microservices on Kubernetes, which call the Mambu-based core banking platform. Side panels hold external integrations, the Sharia compliance engine, the AI/ML platform, and the compliance & audit subsystem. Everything lives on AWS Middle East (UAE).

L1 Client experience EN + AR · WCAG AA · biometric · offline-tolerant

iOS **Mal iOS**

Native Swift. Face ID. UAE Pass deep-link. Push to wake.

Swift SwiftUI SPM

A **Mal Android**

Kotlin / Jetpack Compose. BiometricPrompt. UAE Pass.

Kotlin Compose

W **Mal Web app**

React + TS. Sub-3s LCP. PWA install. RTL-ready.

Next.js React

Op **Ops Console**

CS, Ops, Treasury, Risk, Sharia review queues.

Retool Auth0

mTLS · OAuth 2.1 PKCE · device attestation

L2 Edge · CDN · Security CloudFront UAE PoPs · global WAF · bot mgmt

▲ **CloudFront**

CDN, TLS 1.3, regional edge cache for static assets.

AWS

⌘ **WAF + Shield Advanced**

OWASP top-10 · L7 DDoS mitigation · IP rep lists.

AWS Cloudflare Bot Mgmt

∞ **API gateway**

Envoy + Istio. mTLS, rate limit, schema enforcement.

Istio Envoy

id **Identity (OIDC)**

Auth0 + UAE Pass federation. Step-up auth for risk events.

Auth0 UAE Pass

EXT External integrations

Pluggable adapters, schema-versioned

A **Aani · UAEFTS**

CBUAE instant + RTGS rails. Domestic AED in seconds.

B **Buna · SWIFT**

Arab Monetary Fund + global wires.

V **Visa / Mastercard**

Card schemes for SME spend cards.

N **Nymcard**

MENA-native issuer processor. BIN sponsorship via partner.

K **UAE Pass + Sumsub**

UAE Pass federation; Sumsub for KYB + UBO graphs + liveness.

C **Refinitiv · Dow Jones**

Sanctions · PEP · adverse media · perpetual screening.

E **AECB**

Etihad Bureau API. Mandatory for SME credit decisions.

L **Lean Tech**

First CBUAE IPA holder. 12-mo statements, balances, PIS.

P **Network + Magnati**

POS data feed · daily revenue signal for pre-approved offers.

W **Wafeq · Zoho · Xero · Tally**

Accounting native connectors + Codat for long tail.

REST · gRPC · async events



Avalara · RTC Suite

FTA-accredited Peppol ASPs. Dual integration · no SPOF.



Currencycloud

FX, multi-currency holding, hedging.

SCH Sharia compliance engine

AAOIFI · UAE Higher Sharia Authority · Dar Al Sharia SSB



SSB Console

Dar Al Sharia / Shariyah Review Bureau · fatwa register · quarterly audit.



Murabaha engine

Invoice finance: cost + markup, deferred payment, asset-backed.



Halal screening

Counterparty industry codes blocked at txn-time (IdealRatings).



Late fee → charity

Ta'widh diverted to registered charity. No income to Mal.



Zakat calculator

Hawl tracking, nisab thresholds, payable amount per SME.



UAE HSA submission

Every product structure submitted to Higher Sharia Authority pre-launch.

L3 Backend-for-frontend & domain services

Kubernetes (EKS) · 3 AZ · 99.95% SLO · TypeScript + Go

DE *Decision Engine*

The brain. One engine for underwriting, pricing, EWS, fraud, collections routing, pre-approval. Composes rules, policies, manual steps, referrals, and models. Built in-house.

Build · Day 1

Pr *Pricing Engine*

Risk-tier pricing, dynamic on supply / demand. Tightly coupled with Decision Engine. Built in-house.

Build · Day 1

\$ *Account & balance*

Multi-currency wallets, IBAN map, holds, sweeps.

Build

→ *Payments orchestrator*

Rail selection, idempotency, retries, sagas. Money-movement IP.

Build

LO *LOS*

Loan Origination. Workflow IP, configurable by Risk without engineering. Built in-house.

Build · Day 1

CX *Customer graph*

Onboarding, KYC orchestration, UBO graph, consent, signatories. Canonical record.

Build

□ *Cards & expense*

Card issuance via Nymcard. Per-employee controls, receipt OCR, policy engine: built.

Hybrid

Lo *Murabaha engine*

Invoice financing contract lifecycle. SSB-template-driven, EMIs.

Build

AI *AI-native decisioning platform*

Five layers. One Decision Engine. ~20 agents.

1 *Customer surfaces*

Buyer app, supplier app, anchor portal, ops console.

2 *Agent orchestration*

Tool router, replay log, eval harness, human-override. Temporal + LangGraph.

3 *Agent fleet (~20)*

Origination, underwriting, servicing, ops, risk and CX. Each agent has one job and a HITL trigger.

4 *Models & features*

Feature store, embedding store, managed LLM serving. Model class varies by task.

5 *Data foundations*

AECB · EmaraTax · Open Finance (Lean) · POS (Network + Magnati) · internal events.

DE *Decision Engine*

The brain. Mix of rules, policies, manual, referrals, models. Same engine for every customer-decision the bank makes.

FX sub-ledger

Multi-currency book. FX engine sits behind a vendor; hedging policy is ours.

Hybrid

Portfolio monitor

Real-time book health, EWS dashboards, cohort vintage, regulatory reporting.

Build

Collections platform

Manual at launch. Built in-house as NPL volume scales. EWS is the priority investment.

Build (phased)

Notifications

SMS, push, email, WhatsApp, in-app streams.

Buy

events · CDC · sagas

L4 **Platform plumbing** Event-sourced, workflow-orchestrated, traced end-to-end

Event bus

Kafka (MSK). 7-day hot, 365-day cold replay. AVRO + Schema Registry.

MSK

Avro

Workflow engine

Temporal for sagas: KYC, money movement, Murabaha lifecycle.

Temporal

Feature flags

LaunchDarkly. Per-emirate, per-product, per-customer-segment rollouts.

LaunchDarkly

Object store

S3 with Object Lock for documents, contracts, statements, e-invoices.

S3

WORM

double-entry journal · webhooks

REG **Regulatory & compliance**

ADGM FSP Cat 2+4 · CBUAE · FTA · UAE FIU

ADGM FSP

Cat 2 (Providing Credit) + Cat 4 (Operating PFP). RegLab opt.

CBUAE supervision

Open Finance Circular 3/2025 · Customer Protection (13 Sep 2026).

goAML · SAR

STR + SAR filings to UAE FIU. Real-time AML alerts.

Audit log

Append-only · 10-yr retention · S3 Object Lock.

FTA Peppol

5-corner e-invoice. SME mandate from July 2027 · ready by Q3 2026.

Data residency

UAE-only PII · UAE HSA submission for Sharia products.

L5 *Core banking platform*

Mambu SaaS · multi-tenant on UAE region · 99.99% uptime SLA

Mambu Core

Accounts, deposit products, loan accounts, lifecycle hooks.

SaaS

Double-entry ledger

Append-only journal. Sub-ledger per product. Real-time GL roll-up.

Mambu

Custom GL

IBAN issuer

Partner bank (Wio / RAKBank). Virtual IBAN per SME account.

Wio · partner

CDC streams · daily snapshots

OPS *Observability & SRE*

Three pillars + business SLOs

Datadog

Metrics, APM, logs, RUM, synthetics.

Tracing

OpenTelemetry, sampled at 1% + 100% on errors.

PagerDuty

On-call rotations, follow-the-sun, sev runbooks.

Chaos & DR

Quarterly game-days; warm DR to Bahrain region.

L6 *Data platform*

Polyglot persistence · OLTP + streaming + lakehouse · feature store

PG *PostgreSQL*

Per-service OLTP. Aurora MZ. PITR + 90-day point-in-time.

Aurora

RD *Redis / Valkey*

Session, rate-limit, idempotency, hot feature cache.

Elasticache

CH *ClickHouse*

Analytics OLAP. Real-time merchant insights & dashboards.

CH Cloud

Δ *Lakehouse*

Iceberg on S3, dbt models, Airflow, served via Trino.

Iceberg

dbt

FS *Feature store*

Feast (offline+online). Versioned features for fraud + credit.

Feast

Decision Engine

This is the heart of the platform. Underwriting, pricing, EWS, fraud, collections routing, pre-approval all run through the same five steps. Each step composes some mix of rules, policies, manual workflow, referrals, and models. The model class depends on the question.

step 01 Data pull AECB, e-invoice match, bank feed, KYC docs, POS, prior platform footprint. Parallel where possible, cached behind the feature store. data	step 02 Feature build Canonical features: cashflow ratios, exposure, sector, seasonality, behaviour. Versioned, lineage-tracked. features	step 03 Score Whatever model class fits the question. Behavioural and deep models for acquisition. Scoring families for underwriting. Survival for EWS. Output is a signal, not a decision. models	step 04 Policy gate Deterministic rules. Sharia, sanctions, exposure caps, DSCR floors, KYC pass / fail. Binary, auditable. Overrides the model in either direction. rules
step 05 Price + decide Risk-tier price, limit, plan ladder, customer-facing reason codes (SHAP for the regulator, plain language for the customer). decide	applies to Same engine, many questions Acquisition · onboarding · underwriting · pricing · in-life limit review · EWS · fraud · collections routing. One engine, one trace log, one audit pack. Built in-house from day one. Risk team configures rules without engineering lift. build · day 1		

R ***Rules***

Sanctions, blacklist, minimum eligibility, exposure caps, DSCR floors, sector concentration. Binary. Regulator-defensible.

P ***Policies***

Risk-team-owned, configurable in the engine without engineering lift. Versioned. Approved by the Credit Forum.

M ***Models***

PD, LGD, EWS, fraud, doc verification, propensity, look-alike. Model class chosen by task. SHAP for every decision.

Mn ***Manual***

Any case the AI scores below threshold, any ticket above Express ceiling, any first-time pair above policy, any sector flagged by the SSB.

Rf ***Referrals***

Out-of-policy cases route to named committees with SLAs. Sharia structure to SSB. Cross-product to Credit Forum. Nothing falls through.

******* ***Always on***

Every decision logs input snapshot, model version, policy hash, manual overrides. Queryable by the regulator on request.

Build over buy

The cost of building software has collapsed. The cost of being locked into someone else's product roadmap has not. We build the parts of the stack that touch our customers, our policy, or our IP. We buy where the regulator mandates the counterparty, where the value is genuinely ecosystem-wide (fraud, sanctions), or where the system is a true commodity.

B+ *Build, Day 1*

Decision Engine · LOS · Pricing Engine · Murabaha contract engine · Customer graph · Payments orchestrator · Customer apps · Accounting connectors (5 core) · Onboarding orchestration · Conversational AI (built on managed LLMs) · Portfolio monitoring · EWS · Expense policy engine

core IP

B* *License, then migrate*

LMS (Mambu at launch, then in-house) · Collections platform (manual at launch, then in-house once NPL volume justifies) · FX engine (vendor liquidity, hedging policy ours)

de-risk launch

Bu *Buy*

AECB (mandated) · KYC (Sumsub) · Sanctions / PEP (Refinitiv + Dow Jones) · Fraud intelligence (vendor, hub and spoke) · Commodity broker (DMCC / Bursa) · Card rails (Nymcard) · Open banking (Lean) · Peppol ASP (Avalara + RTC Suite) · Cloud (AWS UAE) · LLM serving (Bedrock) · Observability · EDR

commodity or mandated



Why buy fraud, even though we are build-first

Fraud is hub-and-spoke. A vendor integrated across hundreds of institutions sees a compromised IP, a synthetic identity, or a spoofed device days before any individual lender catches it. That network effect cannot be replicated by building. The vendor sits next to our own fraud signals, not in place of them.



The composability commitment

Every vendor module sits behind a clean API contract from day one. The migration to an in-house module is a backend switch, not a re-architecture. Vendor and in-house can run side by side in shadow mode before cut-over. This is what makes the licence-then-migrate pattern safe.

Money flow

How AED moves when an SME buyer accepts a 60-day invoice using Mal's Sharia-compliant invoice finance. Each column is one timeline step. Read left-to-right per actor row.

<i>ACTOR / TIME</i>	t_0 Invoice issued	t_0+4h Buyer accepts	t_0+4h Mal funds supplier	t_0+30d Buyer EMI 1	t_0+60d Buyer EMI 2 + close
B Buyer SME <i>Crescent Trading FZE</i>	Receives invoice Push from Atlas Packaging AED 250,000	Picks plan 2-EMI · AED 256,500 UAE Pass sign + 3DS	Limit reserved AED 250,000 hold on credit line	EMI 1, AED 128,250 Auto-debit AED account Aani settlement	EMI 2, AED 128,250 Receivable extinguished Contract closes
M Mal <i>As principal</i>	Pre-scores buyer Cached underwriting tier · A	Buys the receivable From Atlas at AED 232,500 (93% advance)	Murabaha contract Sells to buyer Cost AED 250K + AED 6,500 markup	Receives EMI 1 AED 128,250 GL credit · sub-ledger	Receives EMI 2 · profit Net markup AED 6,500 booked
L Mambu ledger <i>Double-entry</i>	JE-0 No movement; offer queued	JE-1 Dr Suspense · Cr Supplier IBAN AED 232,500	JE-2 Dr Murabaha Receivable · Cr Suspense AED 250,000 + AED 6,500 markup acc.	JE-3 Dr Customer AED · Cr Receivable AED 128,250	JE-4 · close Receivable = 0 Profit recognised AED 6,500
R Rails <i>Aani · UAEFTS</i>	·	Aani instant Mal partner IBAN → Atlas IBAN AED 232,500 · <10s	·	Aani Direct Debit Buyer IBAN → Mal IBAN	Aani Direct Debit Final settlement
S Supplier SME <i>Atlas Packaging FZ-LLC</i>	Issues invoice Peppol e-invoice via Wafeq	Awaits offer Mal status: accepted	Wire received AED 232,500 · 4h SLA Holdback AED 17,500	·	Holdback released AED 17,500 · receivable cleared

Note on Sharia structure. Mal does not lend cash at interest. It buys the receivable from the supplier and sells the underlying obligation to the buyer at a disclosed cost-plus markup with deferred settlement (*Murabaha*). The Sharia Supervisory Board's standing fatwa governs the contract template; each transaction's structure is logged and re-affirmed quarterly.

Sharia lifecycle

The eight-stage gate that every invoice-finance request walks through. Yellow gates are decisions; blue gates are actions; green gates create or extinguish an asset.

<i>01 · decision</i> Halal screening Industry-code check on supplier & buyer. Block alcohol, gambling, conventional finance, adult, weapons. SCH · screen	<i>02 · decision</i> Asset existence Invoice references a real, deliverable good or service. Reject services that are pure money-for-money. SCH · asset	<i>03 · action</i> Offer to buyer Cost-plus quote. Discloses Mal's purchase cost, profit margin (markup), tenor, EMI schedule. SCH · disclosure	<i>04 · action</i> Promise to purchase Buyer's binding promise (wa'd). UAE Pass signed. Mal commits to buy from supplier first. SCH · wa'd
<i>05 · asset</i> Mal acquires receivable Mal pays the supplier (Aani). Ownership of the receivable transfers to Mal. JE-1 posted. SCH · own	<i>06 · asset</i> Murabaha sale Mal sells the underlying obligation to the buyer at cost + markup. JE-2 posted. SCH · sell	<i>07 · action</i> EMI collection Deferred-payment instalments. Late fees go to charity (not Mal income). Restructure available. SCH · service	<i>08 · asset</i> Contract closure Receivable extinguished. Profit recognised. Records archived for 10 years; quarterly Sharia audit picks 5% sample. SCH · close



Sharia Supervisory Board

Three external scholars (AAOIFI-certified) + an internal head. Quarterly fatwa review, annual product certification, sample audit on 5% of executed Murabaha contracts.

AAOIFI

Quarterly



Late fees → charity

Penalty income cannot accrue to Mal. Late fees are diverted into a registered charitable account; SMEs see the destination in their statement.

Audit-checked



Product templates

Murabaha invoice-finance template is pre-approved by the SSB · the engine refuses to render a contract outside that template · every clause logged for audit.

Templates

Deployment topology

Primary region: AWS Middle East (UAE) in Abu Dhabi. Three AZs run identical stacks (active-active for stateless, active-passive for stateful with synchronous replication). A warm DR site in AWS Middle East (Bahrain) takes over within a 15-minute RTO and 0-second RPO for journal data.

AWS me-central-1 · Abu Dhabi

PRIMARY · ACTIVE

All customer PII, ledger, journal events, KYC artefacts. CBUAE-compliant data residency.

az-1a

- EKS workers · APIs
- Temporal workers
- Aurora writer
- Redis primary
- MSK broker 1

az-1b

- EKS workers · APIs
- Temporal workers
- Aurora reader
- Redis replica
- MSK broker 2

az-1c

- EKS workers · APIs
- Temporal workers
- Aurora reader
- Redis replica
- MSK broker 3

Shared regional services

S3 · Glacier · KMS · CloudFront UAE PoPs · Direct Connect (Equinix DX1)

External egress

CBUAE EFTS · Aani · AECB · Sumsub · Mambu API · all via PrivateLink or IPSec where supported

AWS me-south-1 · Bahrain

WARM DR

Synchronous Aurora Global, cross-region MSK mirror, EKS clusters at minimum capacity. CBUAE-approved DR location; failover under 15 minutes.

az-1a

- EKS · scaled to 30%
- Aurora Global · reader
- MSK mirror brokers

az-1b

- EKS · scaled to 30%
- Aurora Global · reader
- MSK mirror brokers

RTO 15 min · RPO 0 (for ledger journal) · RPO 30s (for analytics)

Quarterly DR game-day required. CBUAE annual attestation.



KMS · HSM

CloudHSM cluster for card PAN/CVV, signing keys, root CMK; quarterly key rotation.



Egress

All third-party traffic via PrivateLink, VPC endpoints, or IPSec; full TLS 1.3.



Audit replication

All audit logs replicate to Object Lock S3 in both regions; immutable for 10 years.



Cross-border policy

No PII leaves UAE without explicit consent + CBUAE notification; vendor SaaS terms enforced.

Mal · capital that moves with you.

Architecture diagram · v1.0 · 2026-05-13 · Companion to the architecture documentation

Diagrams are normative for system shape, not for vendor SKUs. Replaceable components are explicitly flagged in the documentation.